Anurag Yadav

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SUMMARY

Software Developer specializing in Backend Development, Generative AI, and Agentic AI. Experienced in building RAG systems, hybrid retrieval pipelines, LLM applications, and AI agents using Python, FastAPI, LangGraph, Qdrant, and modern LLM tooling. Strong foundation in Data Structures Algorithms, databases, REST APIs, and scalable backend systems, with hands-on experience in Dockerized deployment and LLM fine-tuning using LoRA/QLoRA.

PROJECTS

RAG-Based E-Commerce Product Assistant [LINK] [VIDEO] *2026*

- Built a hybrid-search RAG chatbot over Amazon product data, combining dense retrieval with OpenAI embeddings and BM25 keyword search in Qdrant using Reciprocal Rank Fusion (RRF), with Instructor-enforced structured LLM outputs and cited product references.
- Developed a ReAct/LangGraph-based e-commerce shopping agent with intent routing and conversation-aware query handling; implemented Cohere reranking after retrieval fusion to improve product relevance.
- Traced the end-to-end RAG pipeline using LangSmith to monitor LLM calls, token usage, latency, and retrieval/generation workflows.
- Developed a FastAPI + React e-commerce app with PostgreSQL-backed cart operations: add to cart, remove from cart, and view cart.
- **Tools Used: Python, FastAPI, Qdrant, OpenAI API, Cohere, LangSmith, React, Docker.**

SketchWaves - Real-Time Collaborative Whiteboard [LINK] *2025*

- Engineered a real-time collaborative whiteboard enabling concurrent multi-user sessions with low-latency canvas synchronization through WebSockets, supporting live drawing, shapes, text, and undo/redo functionality.
- Developed a full-stack application incorporating a React.js frontend, Node.js backend, and MongoDB for persistent session and drawing storage; utilized Docker for scalable and reproducible deployment.
- Implemented WebSocket-based room/session management and real-time scene synchronization for collaborative boards, with MongoDB persistence for board state.
- **Tools Used: React, Express, Tailwind CSS, MongoDB, Node.js, WebSocket, Perfect-Freehand, Roughjs**

Qwen3-4B Mathematical Reasoning Fine-Tuning (QLoRA) [Github] *2026*

- Fine-tuned Qwen3-4B for mathematical reasoning using QLoRA/PEFT, training LoRA adapters while keeping the base model quantized in 4-bit precision.
- Processed 2,000 OpenMathReasoning examples into Qwen chat-template format and tokenized them for supervised causal language-model training.
- **Tools Used: Google Colab, Hugging Face (Transformers, PEFT, Datasets), BitsAndBytes, PyTorch**

TECHNICAL SKILLS

- **Programming Languages:** C++, Python , SQL
- **Frontend:** JavaScript (ES6+), React.js, Tailwind CSS,Redux
- **Backend:** Node.js, Express.js, Typescript, Redis, FastAPI , RESTful APIs.
- **Databases:** MongoDB, PostgreSQL, MySQL, ChromaDB, Qdrant, FAISS.
- **Core Concepts:** OOPs, DBMS, Operating Systems, Computer Networks.
- **DevOps/Tools:** Git, GitHub, Postman, Docker.
- **GenAi / Agentic AI :** LLMs, LangChain, LangGraph, Hugging Face, PEFT, (LoRA/QLoRA), TRL, RAG, Tool Calling, Multi-Agent Systems, Pydantic.

EDUCATION

Bachelor of Technology (B.Tech) - Computer Science and Engineering (Data Science) *Sep 2023 - June 2027*
Galgotias College of Engineering and Technology, Greater Noida, India
Current SGPA: **7.7** *(Till 6th Semester)*

AWARDS & ACHIEVEMENTS

- Solved 1000+ DSA Problems across LeetCode,GFG and Codeforces Leetcode Rating 1800+
- Ranked In Top 2000 LeetCode Weekly Contest